Problem 4

Determine the random variable X starting from the equation $AX = AB$.

Solution

We need to solve for the set (or event) X given that $AX = AB$, where the operation is intersection.

Step 1: Interpret the Equation

The equation $AX = AB$ means:

$$A \cap X = A \cap B$$

We need to find all sets X that satisfy this equation.

Step 2: Analyze What the Equation Tells Us

The equation $A \cap X = A \cap B$ states that the intersection of A with X equals the intersection of A with B.
This means:

- Every element in $A \cap B$ must be in X (so that $A \cap B \subseteq A \cap X$)
- Every element in $A \cap X$ must be in $A \cap B$ (so that $A \cap X \subseteq A \cap B$)

Step 3: Derive Necessary Conditions

Condition 1: $AB \subseteq X$

From $A \cap X = A \cap B$, we know that $A \cap B \subseteq A \cap X$.
For any $x \in A \cap B$, we have $x \in A$ and $x \in B$, and we need $x \in A \cap X$, which means $x \in X$.
Therefore:

$$A \cap B \subseteq X$$

or equivalently:

$$AB \subseteq X$$

Condition 2: $A \cap X \subseteq AB$

From $A \cap X = A \cap B$, we also have $A \cap X \subseteq A \cap B$.
This means that any element in both A and X must also be in B.
In other words: if $x \in A$ and $x \in X$, then $x \in B$.
This is automatically satisfied if $AB \subseteq X$ and we don't add extra elements from A that aren't in B.

Step 4: Find the General Solution

For $A \cap X = A \cap B$ to hold, we need:

$$AB \subseteq X$$

But what about elements in X that are not in A?
If $x \notin A$, then:

- $x \notin A \cap X$ (since $x \notin A$)
- $x \notin A \cap B$ (since $x \notin A$)

So elements outside A don't affect the equation $A \cap X = A \cap B$.
Therefore, X can contain any elements outside A.

Step 5: Complete Solution

The general solution is:

$$X = AB + C$$

where C is any subset of $\bar{A}$ (the complement of A), i.e., $C \subseteq \bar{A}$ or $C \cap A = \emptyset$.

Equivalently:

$$\boxed{X = AB \cup C, \quad \text{where } C \cap A = \emptyset}$$

Step 6: Verification

Let $X = AB \cup C$ where $C \cap A = \emptyset$.

Then:

$$A \cap X = A \cap (AB \cup C) = (A \cap AB) \cup (A \cap C)$$

Since $AB = A \cap B \subseteq A$, we have $A \cap AB = AB$.

Since $C \cap A = \emptyset$, we have $A \cap C = \emptyset$.

Therefore:

$$A \cap X = AB \cup \emptyset = AB = A \cap B$$

This confirms our solution.

Special Cases

Minimal Solution

The minimal solution (smallest possible X) is:

$$X_{\min} = AB$$

This is obtained by taking $C = \emptyset$.

Maximal Solution

The maximal solution (largest possible X) is:

$$X_{\max} = AB \cup \bar{A} = AB + \bar{A}$$

This is obtained by taking $C = \bar{A}$ (all elements not in A).

We can verify: $AB + \bar{A} = (A \cap B) \cup \bar{A} = B \cup \bar{A}$ (by absorption).

Alternative Expression

Since $X = AB \cup C$ with $C \subseteq \bar{A}$, we can write:

$$\boxed{X = AB + \bar{A}D}$$

where D is any arbitrary set (because $\bar{A}D \subseteq \bar{A}$ automatically satisfies $(\bar{A}D) \cap A = \emptyset$).

Even more simply:

$$\boxed{X = AB + \bar{A}}$$

is the most general form, where we understand that this represents the class of all sets containing AB and possibly some elements from $\bar{A}$.

Intuitive Interpretation

The equation $AX = AB$ means:

- Within A, the set X must contain exactly the elements that are in B (i.e., $A \cap X = A \cap B$)
- Outside A, the set X can contain anything

Final Answer

The general solution to $AX = AB$ is:

$$\boxed{X = AB + C, \quad \text{where } C \subseteq \bar{A} \text{ (or } C \cap A = \emptyset)}$$

Equivalent forms:

- $X = (A \cap B) \cup C$ where $C \cap A = \emptyset$
- $X = AB \cup (\bar{A} \cap D)$ for any set D
- Minimal solution: $X_{\min} = AB$
- Maximal solution: $X_{\max} = AB + \bar{A} = B \cup \bar{A}$

Interpretation: X must contain all elements in both A and B, and can contain any elements outside A.